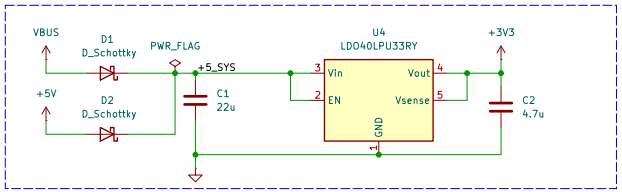
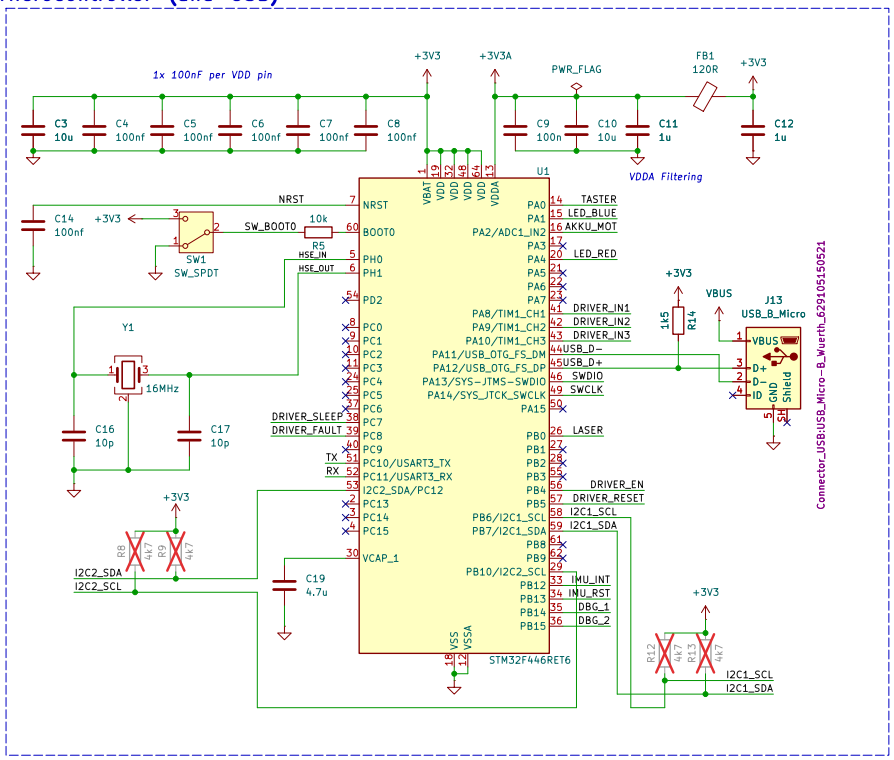
# Prototype 4.5 STM32 PCB

## Power Supply

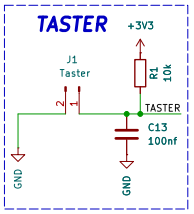



## Microcontroller (and USB)

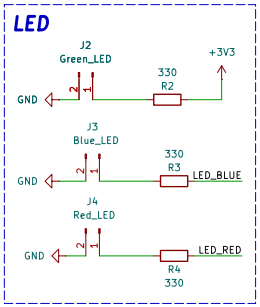



Pin-out using STM32CubeMX

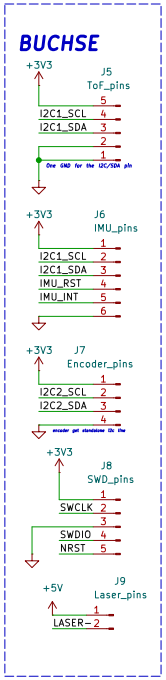
## TASTER



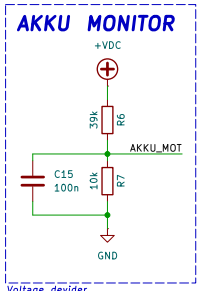

## LED




## BUCHSE

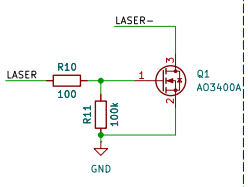



## AKKU MONITOR



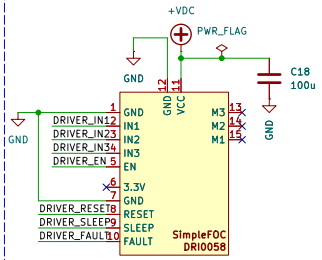

Voltage devider
V_adc = V_akku x (10k/10k + 39k)

## LASER



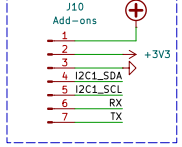

using N-Channel MOSFET as the laser switch

## TREIBER



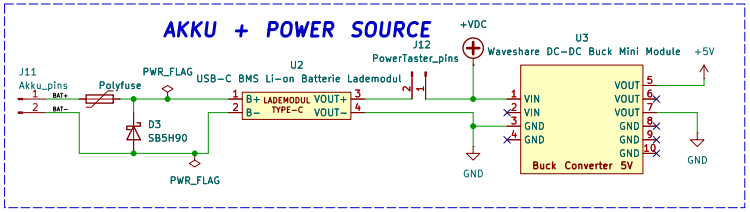

M1,M2,M3 connects directly to the Motor

## ADD-ONS

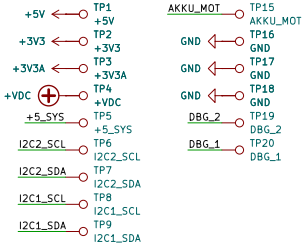



The addon port can be used for hacking the device and adding custom extensions, or while debugging. It exposes 3.3V power, the I2C bus and the RX/TX lines for serial output and communications. Test points provide solderable connections to more GPIO pins as well.

## AKKU + POWER SOURCE




## TEST POINTS




## MOUNTING HOLES




footprint finished, DNP for I2C resistors
adding Test points and mounting holes
adding Laser switch

**FH Aachen**

Sheet: /
File: stm32_kicad.kicad_sch

**Title: Prototype 4.5 STM32 PCB**

| Size: A3 | Date: 2026-10-26 | Rev: 7 |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 1/1 |